

Lavya Midha

Machine Learning Engineer | Applied AI | AI Engineer

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EDUCATION

New York University

Master of Science in Data Science (Industry Concentration, Big Data)

Cumulative GPA: 3.758/4.0

New York, NY

Sept 2024 - May 2026

Boston University

Bachelor of Science in Data Science, Minor: Statistics

Cumulative GPA: 3.75/4.0

Boston, MA

Sept 2021 - May 2024

WORK EXPERIENCE

ReferU.AI

Machine Learning Engineer

New York, NY

May 2025 – Present

- Built a multi-agent legal research platform using LangChain, CrewAI, OpenAI, FastAPI, and GCP that retrieves cases, validates citations, and generates citation-backed legal answers.
- Developed machine learning evaluation pipelines across 10,000+ legal citations, measuring retrieval quality, ranking performance, hallucination rates, and model robustness to improve answer quality by 16%.
- Built backend services for attorney intake and marketplace operations, supporting consultation booking, referral tracking, outcome reporting, and fee management workflows.
- Designed database models, migrations, REST APIs, audit logging systems, and workflow automation to support production legal and marketplace applications.
- Optimized search and retrieval pipelines through ranking improvements, query optimization, and automated evaluation frameworks, increasing system reliability and answer relevance at scale.
- Technologies: LangChain, CrewAI, OpenAI, FastAPI, PostgreSQL, GCP, Docker

Trilemma Foundation

Data Science Consultant

New York, NY

Sept 2025 – Dec 2025

- Built machine learning pipelines integrating time-series forecasting, NLP-based sentiment analysis, and adaptive decision frameworks for investment strategy development.
- Engineered preprocessing workflows across 300+ features using Random Forest MICE imputation, correlation analysis, and variance inflation factor screening.
- Developed automated backtesting and evaluation systems spanning 10+ years of historical data to benchmark model performance and robustness.
- Designed and deployed a Streamlit dashboard providing real-time access to model recommendations, performance analytics, and sentiment insights.

BU Spark!

Project Manager

Boston, MA

Jan 2024 – May 2024

- Led cross-functional teams delivering machine learning and analytics solutions for civic-sector organizations.
- Supported development of predictive models, data pipelines, and Tableau dashboards for stakeholder reporting.

Boston University School of Public Health

Data Science Research Intern

Boston, MA

Feb 2023 - Dec 2023

- Developed time-series forecasting and anomaly detection models on environmental sensor data collected across 125+ schools, enabling proactive infrastructure and operational planning.
- Built OpenCV-based computer vision pipelines to extract spatial features from architectural floor plans.
- Analyzed large-scale environmental datasets to support infrastructure planning and public health research initiatives.
- Designed real-time anomaly detection dashboards, reducing operational response times by 35%.

SKILLS

Programming: Python, SQL, C++, R

Machine Learning: PyTorch, TensorFlow, XGBoost, Scikit-learn, Deep Learning, Recommendation Systems, Reinforcement Learning, Natural Language Processing, Computer Vision, Time Series Forecasting, Feature Engineering, Model Evaluation, A/B Testing

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Vector Search

Backend & Infrastructure: FastAPI, REST APIs, PostgreSQL, Docker, Apache Spark, PySpark, Hadoop, ETL Pipelines, CI/CD, Git, GitHub

Cloud: GCP, AWS

Data & Visualization: Pandas, NumPy, Tableau, Power BI